INCH-POUND

MIL-DTL-32022A(OS)
23 July 2025
SUPERSEDING
MIL-DTL-32022(OS)
5 November 1998

DETAIL SPECIFICATION

GENERATOR, GAS, FIRE EXTINGUISHER, GGU-16/A, GGU-17/A, GGU-18/A, GGU-19/A, GGU-20/A

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the GGU-16/A, GGU-17/A, GGU-18/A, GGU-19/A, and GGU-20/A generator, gas, fire extinguisher, referred to herein as "gas generator."

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

MIL-DTL-23659 - Initiators, Electric, General Design Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS (CAGE CODE 30003)

3205AS156 - Generator, Gas Fire Extinguisher GGU-19/A

3205AS157 - Generator, Gas Fire Extinguisher GGU-18/A

3205AS158 - Generator, Gas Fire Extinguisher GGU-16/A

3205AS159 - Generator, Gas Fire Extinguisher GGU-17/A

3205AS160 - Generator, Gas Fire Extinguisher GGU-20/A

3205AS191 - Test Fixture Assembly (100±0.5 Liter)

(Copies of these drawings are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

BELL-BOEING DOCUMENTS

901-910-034 - V-22 Data EMD Equipment Vibration, Shock, Acceleration, and Acoustics Test Plan

901-947-222 - V-22 Data Wing Fire Detection/Suppression System

(Application for copies of Bell-Boeing documents should be sent to Bell Helicopter Textron Inc., 600 E Huyrst Blvd, Fort Worth, TX 76101.)

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this document are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS-STD-595 - Colors Used in Government Procurement

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.3).

3.2 Flight critical part. These gas generators are flight critical items (see 6.4.2), and the technical data package shall be identified and procured as such.

3.3 Design and construction. The GGU-16/A, GGU-17/A, GGU-18/A, GGU-19/A, and GGU-20/A gas generators covered by this specification shall be manufactured in accordance with Drawings 3205AS158, 3205AS159, 3205AS157, 3205AS156, and 3205AS160, respectively.

3.3.1 Materials. All materials used in the manufacture of the gas generators shall conform to the standards established at technical service release as specified (see 6.2). Any divergence shall require technical evaluation and potential design certification testing.

3.4 Primary components. The electro-explosive device (EED) and propellants are primary components (see 6.4.3). Only one lot of each primary component shall be used in a gas generator lot. One primary lot may be used in more than one gas generator lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

3.4.1 Electro-explosive devices (EED). The zirconium potassium perchlorate ignition charge and the titanium hydride/potassium perchlorate main charge are the EED's primary components. Only one lot of each primary component shall be used in an EED lot. One primary lot may be used in more than one EED lot. The EED shall be in accordance with the requirements of MIL-DTL-23659.

3.5 Gas generator production.

3.5.1 Visual requirement. The gas generators shall conform to Drawings 3205AS158, 3205AS159, 3205AS157, 3205AS156, and 3205AS160 as applicable and be free of the following defects: illegible, missing, or inaccurate markings; damage, burrs, dents, or sharp edges; bent or missing electrical contact pins; damaged mounting studs or cushions; chipped or incomplete protective finish; or other defects that may prevent the installation of the gas generators.

3.5.1.1 Shorting plug. The electrical initiator shall have a shorting plug installed that shorts all pins to case.

3.5.1.2 Protective caps. Protective caps (see 6.4.1) shall be installed on the electrical connector and mounting studs.

3.5.1.3 Paint. The gas generators shall be painted white (FS-17925) in accordance with SAE AMS-STD-595.

3.5.2 Radiographic requirement. When radiographically examined in accordance with 4.5.2, the gas generator shall show proper assembly, presence of parts, uniformity of propellant pellets, sealing, and proper assembly of the electrical initiator. The radiographic inspection shall show no foreign materials present.

3.5.3 Leakage. Gas generators shall not exhibit a leak rate of 1.0×10^{-5} cubic centimeters per second (cm^3/s) of air or 2.7×10^{-5} cm^3/s of helium when tested in accordance with 4.5.3.

3.5.4 Bridgewire resistance. The bridgewire resistance shall be 1.0±0.1 ohms at 70 °F between pins "A" and "C" when tested in accordance with 4.5.4.

3.5.5 Insulation resistance. The insulation resistance shall be greater than 2.0 megaohms when measured between shunted pins "A" and "C" and the gas generator electrical bonding surface when tested in accordance with 4.5.5. For GGU-16/A, GGU-17/A, GGU-18/A, and GGU-20/A, the bonding surface shall be the mounting stud closest to the electrical initiator. For GGU-19/A, the electrical bonding surface shall be the electric initiator body.

3.6 Environmental.

3.6.1 Vibration. The gas generator shall not initiate or incur any internal/external damage or degradation when vibration cycles are applied in accordance with 4.5.6.

3.6.2 Shock. The gas generator shall not initiate or incur any internal/external damage or degradation when saw tooth shocks are applied in accordance with 4.5.7.

3.6.3 Temperature-shock/humidity/altitude (TSHA) cycling. The gas generator shall not initiate or incur any internal/external damage or degradation when subjected to TSHA cycling in accordance with 4.5.8.

3.6.4 Salt fog. The gas generator shall not initiate or incur any internal/external damage or degradation when subjected to salt fog testing in accordance with 4.5.9.

3.6.5 Six-foot drop. The gas generator shall not initiate or incur any major internal/external damage or degradation as a result of the six-foot drop test in accordance with 4.5.10.

3.6.6 Thermal-shock cycling and vibration. The gas generator shall not initiate or incur any internal/external damage or degradation when subjected to the combined thermal-shock cycling and vibration in accordance with 4.5.11.

3.7 Functional. The gas generator shall meet the requirements of 3.7.1 through 3.7.7 when fired in accordance with 4.5.12.

3.7.1 Generator discharge time. The gas generators shall be completely discharged within 250 milliseconds maximum from the time of initiator actuation.

3.7.2 Time to 95 percent of maximum pressure. The limit for time to 95 percent of the maximum pressure shall not exceed 200 milliseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

3.7.3 Ignition delay time. The ignition delay shall not exceed 25 milliseconds. The ignition delay shall be from current application to the first indication of pressure.

3.7.4 Pressure limits. The upper and lower pressure limits for maximum pressure (psia) when function at the three specified temperatures shall be as follows in table I:

TABLE I. Pressure limits.

	Pmin (psia)	Pmax (psia)
GGU-16/A	23	45
GGU-17/A	58	105
GGU-18/A	11	21
GGU-19/A	20	40
GGU-20/A	48	80

3.7.5 Delta weight. The minimum delta weight of the gas generators following functioning shall be recorded for information only. The change in weight shall not be less than listed in table II.

TABLE II. Delta weight.

	Delta Wt. –65 °F (grams)	Delta Wt. +70 °F (grams)	Delta Wt. +160 °F (grams)
GGU-16/A	120	124	126
GGU-17/A	236	244	254
GGU-18/A	60	62	64
GGU-19/A	110	112	114
GGU-20/A	198	200	206

3.7.6 Initiator all-fire current. The electrical initiator shall function the gas generator when provided with a direct current of 3.5 amperes for 10 milliseconds.

3.7.7 Case temperature. The case temperature shall be measured and shall not exceed +700 °F over the functional temperature range as defined in 4.5.12.5. When the case temperature exceeds 700 °F, reigniting fuel spray or hydraulic fluid will likely occur in the aircraft.

3.8 Workmanship. The gas generators shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes, sealing, and assembly operations when inspected in accordance with 4.5.1.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Production inspection (see 4.3).

c. Lot acceptance inspection (see 4.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

4.2 First article inspection. Unless otherwise specified (see 6.2), a first article inspection sample of 36 gas generators conforming to each drawing 3205AS156 through 3205AS160 shall be subjected to first article testing. Thirty-three of these gas generators shall be expended in the tests listed in table III and three shall be retained for investigative purposes. Any damage inflicted by the environmental exposures that would adversely affect the performance of the item in service application shall be cause for rejection of the first article sample. Any production prior to notification by the contracting agency of the first article sample acceptability is at the contractor's risk. First article testing is for the purpose of determining that the contractor, prior to starting production, is capable to producing items that comply with the technical requirements as specified herein. A first article inspection shall be required if two or more years have passed since the item was manufactured, after major turnover in personnel, following major changes to the production facility, or following transfer of production to a new facility or site. Failure of any gas generator to comply with the requirements of table III shall be cause for rejection of the first article represented.

TABLE III. First article test plan.

Test sequence	Test paragraph	Requirement paragraph	Sample groups								
			I	II	III	IV	V	VI	VII	VIII	IX
1. Visual	4.5.1	3.5.1	6	3	3	3	3	3	3	3	3
2. Radiographic	4.5.2	3.5.2	6	3	3	3	3	3	3	3	3
3. Leakage	4.5.3	3.5.3	6	3	3	3	3	3	3	3	3
4. Bridgewire resistance	4.5.4	3.5.4	6	3	3	3	3	3	3	3	3
5. Vibration	4.5.6	3.6.1	6	---	---	---	---	---	---	---	---
6. Shock	4.5.7	3.6.2	---	3	---	---	---	---	---	---	---
7. TSHA	4.5.8	3.6.3	---	---	3	---	---	---	---	---	---
8. Salt fog	4.5.9	3.6.4	---	---	---	3	---	---	---	---	---
9. 6-foot drop	4.5.10	3.6.5	---	---	---	---	3	---	---	---	---
10. Thermal shock cycling/vibration	4.5.11	3.6.6	---	---	---	---	---	3	---	---	---
11. Function test	4.5.12	3.7	---	---	---	---	---	---	---	---	---
-65 °F±5 °F			2	1	1	1	1	1	3	---	---
70±5 °F			2	1	1	1	1	1	---	3	---
160±5 °F			2	1	1	1	1	1	---	---	3

4.3 Production inspection. All production gas generators shall be inspected and screened for defects. Gas generators failing to meet the requirements listed in table IV shall be rejected and removed from the lot.

TABLE IV. Production inspection.

Test sequence	Test paragraph	Requirement paragraph
1. Visual	4.5.1	3.5, 3.5.1, 3.7
2. Radiographic inspection	4.5.2	3.5.2
3. Leakage	4.5.3	3.5.3
4. Bridgewire resistance	4.5.4	3.5.4
5. Insulation resistance	4.5.5	3.5.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

4.4 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests specified in table V. Failure of any sample gas generator to comply with the requirements listed in table V shall be cause for rejection of the lot represented.

TABLE V. Lot acceptance inspection and tests.

Inspection/test	Test paragraph	Requirement paragraph	Quantity
1. Visual inspection	4.5.1	3.4, 3.5.1, 3.7	Test and retain sample
2. Radiographic inspection	4.5.2	3.5.2	Test and retain sample
3. Leakage	4.5.3	3.5.3	Test and retain sample
4. Bridgewire resistance	4.5.4	3.5.4	Test and retain sample
5. Insulation resistance	4.5.5	3.5.5	Test and retain sample
6. Function test:	4.5.12	3.7	---
-65±5 °F	---	---	1/3 sample
70±5 °F	---	---	1/3 sample
160±5 °F	---	---	1/3 sample

4.4.1 Sample size. Unless otherwise specified (see 6.2), a random sample of gas generators from each production lot, including the samples retained for investigative purposes, shall be selected in accordance with table VI for lot acceptance inspection. Test sample gas generators and samples retained for investigative purposes shall not be applied as part of the quantity specified for delivery (see 6.2).

TABLE VI. Lot acceptance sampling.

Lot size	Test sample size	Retained sample size
2-50	5	3
51-90	6	3
91-150	10	3
151-250	15	3
251-500	21	3

4.5 Inspections and tests.

4.5.1 Visual inspections. The external condition and appearance of the gas generators shall be determined by comparison to Drawings 3205AS156 through 3205AS160. Specific inspections of the electrical connector, shunting clip, mounting studs, aluminum case, and labels shall be conducted at a minimum. Each gas generator shall meet the requirements of 3.5.1 and 3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

4.5.2 Radiographic examination. Radiographic examination shall be conducted in accordance with ASTM E1742/E1742M. All gas generators shall be identified with serial numbers prior to examination. The gas generators shall be arranged on boards or trays in consecutive order with any missing serial numbers identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4- x 6-inch region maximum. The radiographic identification shall include the drawing number, the complete lot number, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the lot acceptance tests (see 6.2). Any observable imperfections as outlined in 3.5.2 shall be cause for rejection of the gas generator. Defective gas generators found during radiographic review are to be marked on the radiographic plate and removed from the production lot. Radiographic equipment settings shall be noted on the radiographic review sheets.

4.5.3 Leakage test. The gas generators shall be leak tested by flushing the loaded gas generator with helium at a pressure of 1+0.1/-0.0 atmospheres absolute, sealing it with the installation of the electric initiator. The helium filled gas generator shall then be washed with dry nitrogen and placed in a vacuum chamber and checked for leakage at one atmosphere differential pressure. The gas generators shall meet the requirements of 3.5.3.

4.5.4 Bridgewire resistance. The initiator bridgewire resistance shall be measured across pins “A” and “C” at 70 °F. The test current shall not exceed 50 milliamperes. The bridgewire resistance shall meet the requirements of 3.5.4.

4.5.5 Insulation resistance. The insulation resistance of the gas generators shall be measured by installing a shunt between pins “A” and “C” and using a megaohm meter, applying 500±10 volts of direct current between the shorted initiator pins and the initiator body for a minimum of 2 minutes. The insulation resistance shall meet the requirements of 3.5.5.

4.5.6 Vibration. The gas generators shall be subjected to the vibration requirements of Bell-Boeing report 901-910-034 as follows: 2 units at -65 °F, 2 units at 70 °F, and 2 units at 160 °F. Each gas generator shall be subjected to the combined source dwell and random excitation environments or individual source dwell and random excitation environments as applicable at -65, +70, and +160 °F in each axis for 20 minutes. A change in gas generator resonant properties following exposure to the source dwell shall constitute a test failure even if the gas generator performance is unaffected.

a. Conduct a resonance survey for each gas generator configuration in each of the three mutually perpendicular axes both prior to and following exposure to the source dwell environment. Conduct a low level (0.5 g peak), 5-500 hertz sweep at a frequency rate of ½ octave per minute (15 minutes duration) with a tolerance equal to ±10 percent on the vibration amplitude and ±2 percent or 1 hertz, for determination of the gas generator configuration’s resonant frequency and transmissibility. The same gas generator serial number of each configuration and same mounting location on the vibration fixture shall be utilized for both the pre- and post-source dwell frequency survey. The resonant frequency shall be considered to be any natural frequency with an amplification factor of 2.0 or greater output response relative to the input level. The gas generator response shall be measured on the housing at the least supported location, or a location shown to have significant amplification. All resonance surveys shall be conducted at ambient temperature. Following the completion of the source dwell vibration testing at -65 °F and 160 °F, the gas generators shall be returned to ambient temperature prior to completion of the post-source dwell resonance survey.

b. Predicated upon the outcome of the resonance survey, source dwell vibration and random vibration excitation are to be conducted simultaneously. If based upon the resonance survey, the four Government desired test frequencies exceed the permitted test frequencies and the source dwell vibration and random excitation cannot be applied simultaneously, the source dwell vibration levels shall be increased by 0.5 g’s.

Source dwell vibration shall be performed at each of the four amplitudes and frequencies established by the Government based upon the resonance survey along each of the three mutually perpendicular axes. Source dwell input levels are shown on figure 1 and shall include the superimposed broad band random background levels of the random excitation as described in item c.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

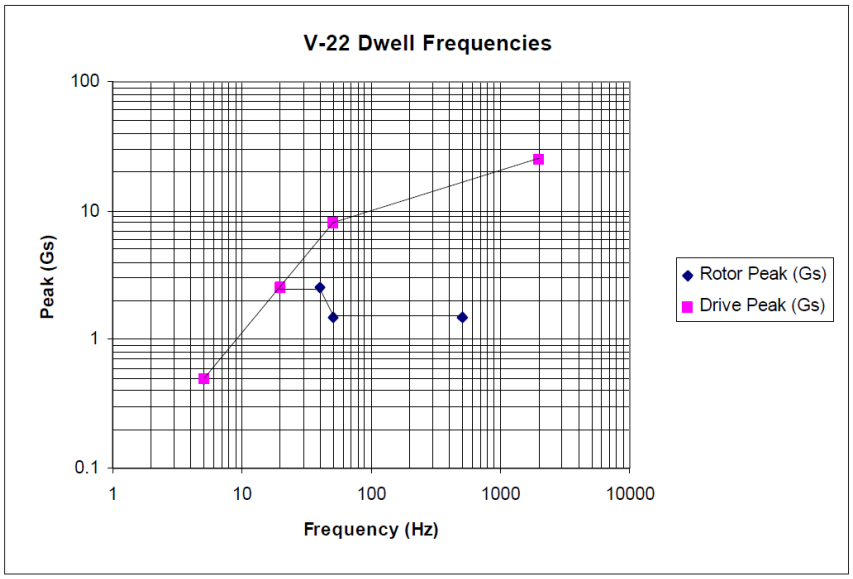


FIGURE 1. Source dwell input levels.

c. Random excitation shall be conducted in accordance with MIL-STD-810, Method 514, Technique 3. Random excitation shall be a flat spectrum at 0.02 g^2/hertz over the bandwidth from 10-2000 hertz as shown on figure 2.

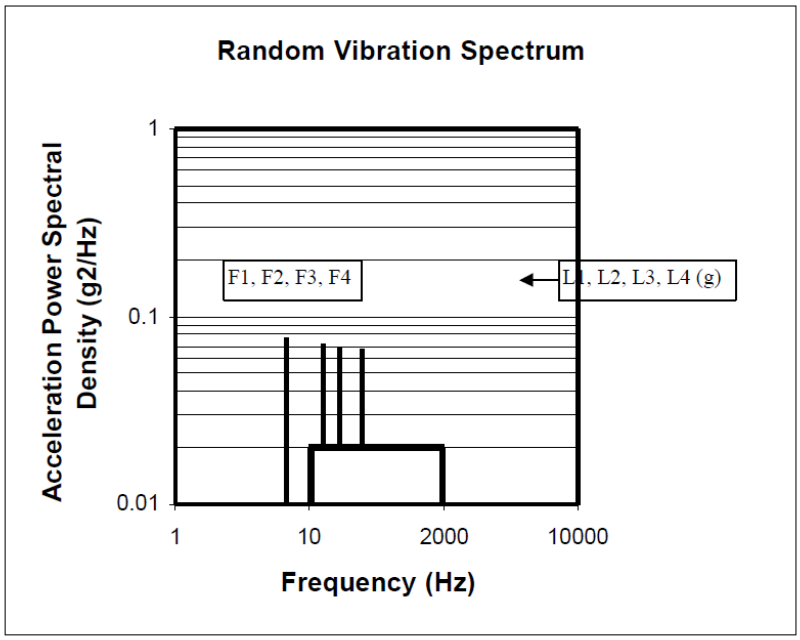


FIGURE 2. Random vibration spectrum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

d. The following tolerances apply to all vibration testing:

(1) Instantaneous random vibration acceleration peaks shall be limited to three times the RMS acceleration (3 sigma clipping).

(2) The random vibration spectrum shall be within the tolerance range of ±3 decibels between 10 and 2000 hertz except that deviations of ±6 decibels are permissible over a cumulative bandwidth of 200 hertz maximum between 500 and 2000 hertz.

(3) Overall RMS acceleration levels shall be within ±1.5 decibels of the nominal input level.

(4) Sine tones shall be within ±10 percent of the nominal value.

e. Failure of any gas generator to comply with the requirements of 3.6.1 shall be cause for rejection of the first article.

4.5.7 Shock. The gas generators, with protective caps and shunts installed on the electric initiators shall be subjected to the shock test outlined in MIL-STD-810, Method 516, Procedure I to the levels in figure 3, with 3 shocks in each direction along each of the 3 axes for a total of 18 shocks. All shocks shall be conducted at ambient temperature. Failure of any gas generator to comply with the requirements of 3.6.2 shall be cause for rejection of the first article.

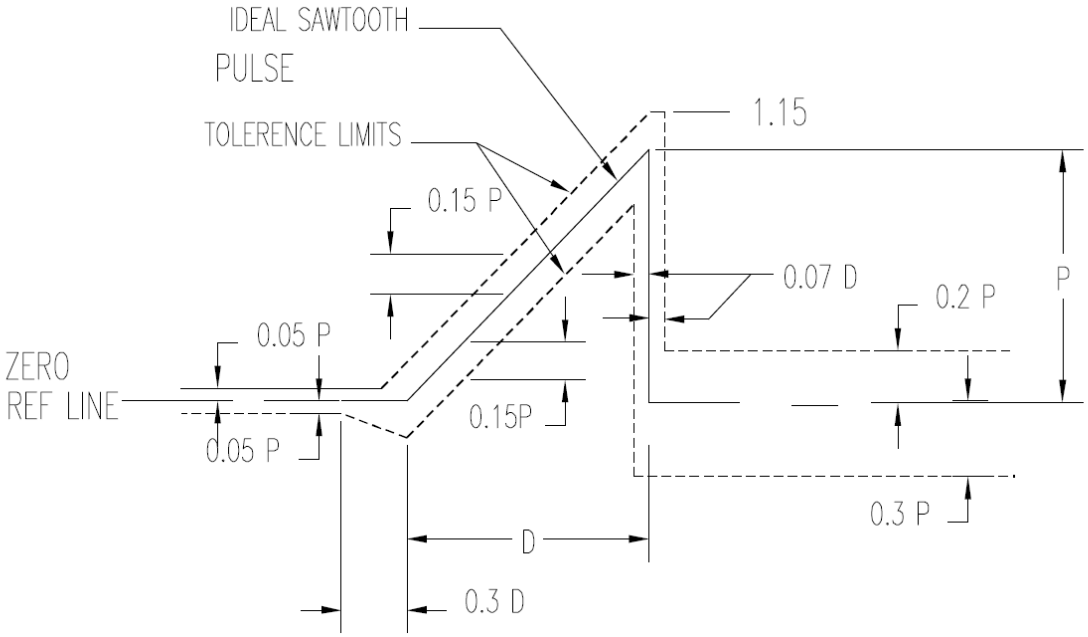


FIGURE 3. Terminal peak saw tooth shock pulse.

4.5.8 TSHA cycling. The gas generators with protective caps installed on the electrical connector shall be subjected to the TSHA cycling test outlined in MIL-D-21625, with the following requirements:

a. The gas generators shall be supported in such a way that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test. No gas generator shall be in contact with another gas generator during the cycling schedule shown in table VII.

b. Failure of any gas generator to comply with the requirements of 3.6.3 shall be cause for rejection of the first article.

c. Thirty thousand (30,000) feet shall be used in lieu of 70,000 feet for the altitude test.

d. There shall be no interim withdrawal of units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

TABLE VII. Temperature shock/humidity/altitude cycling schedule.

Day	Time	Procedure
Monday	0800	Place gas generators (GGs) in chamber at 70 °F at 50% relative humidity (RH).
	1200	Raise chamber temperature to 160 °F and a RH of 95%. The chamber shall reach 160 °F no later than 1300.
	1600	Remove the GGs from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 30,000 feet (4.36 psi).
Tuesday	0800	Remove GGs from chamber and immediately place in chamber maintained at 70 °F at 50% RH.
	1200	Remove the GGs from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 30,000 feet (4.36 psi).
	1600	Remove GGs from above chamber and immediately place in a chamber maintained at 160 °F at 95% RH.
Wednesday	0800	Reduce chamber temperature to 70 °F and RH to 50%. The chamber temperature shall reach 70 °F at 50% RH no later than 0900.
	1200	Raise chamber temperature to 160 °F and a RH of 95%. The chamber shall reach 160 °F no later than 1300.
	1600	Remove the GGs from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 30,000 feet (4.36 psi).
Thursday	0800	Remove GGs from chamber and immediately place in chamber maintained at 70 °F at 50% RH.
	1200	Remove the GGs from above chamber and immediately place in a chamber maintained at -65 °F at a pressure altitude of 30,000 feet (4.36 psi).
	1600	Remove GGs from above chamber and immediately place in a chamber maintained at 160 °F at 95% RH.
Friday	0800	Reduce chamber temperature to 70 °F and RH to 50%. The chamber temperature shall reach 70 °F at 50% RH no later than 0900.
	1200	Raise chamber temperature to 160 °F and a RH of 95%. The chamber shall reach 160 °F no later than 1300.
	1600	Remove the GGs from above chamber and immediately place in a chamber maintained at -65 °F at standard ambient pressure.
NOTE: This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of 160 °F at 95% RH.		

4.5.9 Salt fog. The gas generators, with protective caps installed over the electrical connectors, shall be subjected to the salt fog test with the following requirements:

a. Waxed string shall be used to support the gas generators in the chamber. The gas generator shall be oriented in the connector up position. No change in orientation is required during the test. The gas generators shall be exposed to a concentration of 5 percent salt corrosive agent for a constant wetting duration of 168 hours at 95 °F (35 °C).

b. Condensed fog shall be collected at 2 points every 12 hours and the specific gravity, concentration, and pH of the solution measured. After completing the 168 hours of testing, the salt deposits, if any, shall be removed, if necessary, by a gentle washing with water not warmer than 100 °F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

c. After completing the washing, the units shall be stored for a 48-hour drying period at 70 °F. After 48 hours, a visual inspection shall be conducted. Any signs of damage or degradation shall be noted.

d. The failure of any gas generator to comply with the requirements of 3.6.4 shall be cause for rejection of the first article.

4.5.10 Six-foot drop. The gas generators, with protective shipping caps removed, shall be subjected to the drop test outlined in MIL-D-21625 with the following requirements:

a. The gas generators shall be dropped from a height of 6 feet onto a 2-inch-thick steel plate supported by reinforced concrete. One gas generator shall be used for each drop orientation. All impact orientations are shown on figure 4.

b. After each gas generator is dropped, a visual inspection shall occur to document any external damage that occurred and to verify non-initiation of the gas generator.

c. If it is determined that the damage sustained by any unit caused it to be unsafe for handling all testing shall be stopped.

d. Failure of any gas generator to comply with the requirements of 3.6.5 shall be cause for rejection of the first article.

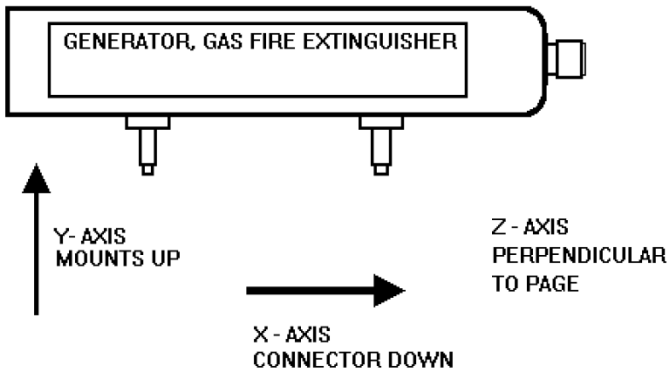


FIGURE 4. Six-foot drop orientations.

4.5.11 Thermal-shock cycling and vibration. The gas generator shall be subjected to the thermal shock cycling and vibration test outlined in Bell-Boeing Report 901-947-222 and MIL-STD-810, Method 503, Procedure I with the following requirements:

a. An option to testing assembled gas generators is to load the propellant tablets into glass vials for the thermal shock cycling, then load the cycled propellant into the gas generators for vibration at the completion of the thermal cycling. The glass vials reduce the thermal soak time due to reduced mass from elimination of generator housing.

b. Stabilize vials at the specified low temperature for the specified range (-65 °F to +10 F, +65 °F to +160 °F, and +95 °F to +190 °F).

c. Increase the ambient temperature to the specified upper limit of the specific range over a 2 minute period.

d. Stabilize at the specified upper limit for 1 minute past thermal stabilization for a gas generator to stabilize.

e. Decrease the temperature to the specified lower limit over a 2 minute period.

f. Repeat this cycle for 168 cycles from -65 °F to 10 °F, 1,344 cycles from 65 °F to 160 °F, and 168 cycles from 95 °F to 190 °F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

g. At the completion of the thermal cycling test the gas generators shall be subjected to vibration testing per 3.6.1.

h. At the completion of thermal shock cycling and vibration testing the gas generators shall be x-ray inspected to the requirements of 3.5.2. Indications of broken pellets shall be cause for halting the test. Gas generators with indications of fractured, broken, or granular pellets shall not be functioned because of the risk of detonation.

4.5.12 Functional performance. The gas generators shall be conditioned for not less than 4 hours and no more than 24 hours at the specified temperature. The gas generators shall be removed from the conditioning chamber and fired in the test fixture 3205AS191 within 5 minutes. If any gas generator is not fired within the 5-minute time period after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested. The gas generator shall meet the performance requirements of 3.7.

a. The GGU-19/A shall be attached to the generator mounting plate with band clamps when installed in the test fixture 3205AS191. The gas generator shall be clamped at the ends of the unit. The band clamp shall be positioned so that pressure vent holes are not blocked and the label is not covered. The other generators shall be attached to the mounting plate using the mounting stud of the generator and wing nuts.

b. After the gas generator is installed, a vacuum pump shall be used to reduce the chamber pressure to 0.5 psia. An absolute pressure gauge shall be used for measuring tank pressure. The tank temperature shall be 70±5 °F before the gas generator is fired.

c. The gas generators shall be initiated by a 5±0.5-ampere pulse of 100 millisecond duration. The gas generators shall meet the requirements of 3.7.1 through 3.7.7.

4.5.12.1 Discharge time. The gas generator discharge time shall be identified as the time between initiator actuation and the time at 95 percent of maximum pressure. The discharge time shall be less than 250 milliseconds.

4.5.12.2 Ignition delay. Ignition delay shall be measured from the application of power until first indication of pressure in the 100-liter test fixture.

4.5.12.3 Maximum pressure. The maximum pressure shall not vary outside the pressure range specified for the designated temperature as specified in 3.7.4.

4.5.12.4 Delta weight. The gas generator shall be weighed pre and post functional testing. The change in weight for each unit shall be compared to the requirements specified in 3.7.5.

4.5.12.5 Case temperature. The case temperature measured over the time interval between initiator actuation and 2 minutes shall be as specified in 3.7.7.

4.5.12.6 Test failure. If a test failure is attributable to an assigned cause, excluding the gas generator, the test shall be declared a no-test and repeated.

4.6 Acceptance criteria. The gas generators shall meet the requirements of 3.6 when tested as specified in 4.5.3 through 4.5.12.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These gas generators are intended to be part of the Wing Fire Protection System installed in the MV-22, CV-22, and HV-22 variant aircraft. They are part of an automatic fire detection/suppression system designed to extinguish a fuel fire in the V-22 wing area composed of the midwing area, aft coves, tip rib bays, and areas in-board of the wing feed tanks. Upon detection of a fire by any of the optical detector the generators can be singly discharged or discharged as part of an inter-connected system. The Wing Fire Protection Systems are only used in military aircraft, therefore, the cartridge has no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and of the source control drawing.

b. That the safety precaution requirements of the "DoD Contractor's Safety Manual for Ammunition and Explosives," DoD Manual 4145.26 are applicable. (Note: When this specification is used as a part of the description of work to be accomplished by a Government activity, the safety precautions requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.)

c. Configuration of gas generator required (see 2.2.2).

d. Whether a first article inspection is required and if so, specify the test activity (see 3.1 and 4.2).

e. Materials required as established by the technical service release (see 3.3.1).

f. Production lot size (quantity) and test facility (see 4.4.1).

g. Directions for shipping radiographs of entire production lot and marriage log (report indicating cartridge serial number to initiator serial number), along with the ballistic sample to the activity conducting the production lot acceptance tests (see 4.5.2).

h. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offeror(s) whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents of the first articles. Invitation(s) for bid(s) should provide that the U.S. Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the U.S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Caps. Plastic shipping caps are installed on the end of the mounting studs and over the EED.

6.4.2 Flight critical part. A flight critical part is a part in which a single failure during any operating condition could cause the loss of aircraft or one of its major components, loss of control, or may cause significant personnel injury, including during flight.

6.4.3 Primary components. Primary components are all components in which a functional failure would result in a misfire or malfunction of the gas generator.

6.5 Change provisions.

6.5.1 Thermal-shock cycling and vibration testing. In the event of a major change to facilities (new facility, transfer to new location, etc.) or large turn-over in staff, the government may require the performance of the thermal-shock cycling and vibration test on the propellant to ensure it meets the requirements and performance of the propellant as qualified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.

6.5.2 Changes to configuration. These gas generators were developed to function in specific V-22 midwing area bays. The requirements address air flow into and out of these bays. Of major importance is the over pressure limits of each bay. Changes to the gas generators resulting in increased performance, faster burn times, or greater pressure output could result in the structural failure of wing components with the potential for catastrophic loss of the aircraft. Any changes affecting the ballistic functions of these gas generators will require the approval of Naval Air Systems Command, AIR-4.3.5, and PM-275, and the Bell-Boeing Joint Program Office.

6.6 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.55 should be used to clarify terms.

6.7 Subject term (key word) listing.

Initiator
Propellant

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2025-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:20Z
Check the source to verify that this is the current version before use.